SQL Analysis Report

This report summarizes the main business insights obtained through SQL analysis of the Italian Restaurant Analytics project.

The analysis covers sales, promotions, payment methods, employees, and products using data stored in a SQLite database.

In [1]:
```
# Import the libraries required to connect to SQLite and display query results

import pandas as pd
import sqlite3
```

Database Connection

The report connects to the SQLite database created from the cleaned restaurant datasets.

In [2]:
```
# Connect to the SQLite database

conn = sqlite3.connect("../sql/restaurant_analytics.db")

print("Database connection created successfully.")
```

```
Database connection created successfully.
```

Sales Analysis

This section summarizes the main sales results obtained through SQL queries, including total sales, sales by category, and monthly performance.

Total Sales

The following SQL query calculates total sales during the analyzed period.

In [4]:
```
# Execute the SQL query for total sales

query = """
SELECT
    ROUND(SUM("Sales"), 2) AS Total_Sales
FROM sales;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[4]:

	Total_Sales
0	1344807.28

Key Insight

Total sales during the analyzed period reached $1,344,807.28.

Total Sales by Category

The following SQL query calculates total sales for each business category.

In [3]:

```
# Execute the SQL query for total sales by category

query = """
SELECT
    "Category",
    ROUND(SUM("Sales"), 2) AS Total_Sales
FROM sales
GROUP BY "Category"
ORDER BY Total_Sales DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[3]:

	Category	Total_Sales
0	Food	948232.21
1	Liquor	211747.70
2	Beverages	157079.85
3	Events	27747.52

Key Insight

Food was the dominant sales category during the analyzed period, generating $948,232.21 in total sales and representing approximately 70.51% of overall revenue.

Monthly Sales

The following SQL query summarizes total sales by month.

In [5]:

```
# Execute the SQL query for monthly sales

query = """
```

```
SELECT
    strftime('%Y-%m', "Date") AS Month,
    ROUND(SUM("Sales"), 2) AS Total_Sales
FROM sales
GROUP BY Month
ORDER BY Month;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[5]:

	Month	Total_Sales
0	2025-03	103876.35
1	2025-04	86374.69
2	2025-05	102869.21
3	2025-06	100154.40
4	2025-07	100600.17
5	2025-08	98279.11
6	2025-09	91959.74
7	2025-10	102360.35
8	2025-11	114970.30
9	2025-12	131135.53
10	2026-01	98768.26
11	2026-02	98651.27
12	2026-03	114807.90

Key Insight

Monthly sales varied throughout the analyzed period, with December 2025 recording the highest total sales.

Highest-Sales Month

The following SQL query identifies the month with the highest total sales.

In [6]:
```
# Execute the SQL query for the highest-sales month

query = """
SELECT
    strftime('%Y-%m', "Date") AS Month,
    ROUND(SUM("Sales"), 2) AS Total_Sales
```

```
FROM sales
GROUP BY Month
ORDER BY Total_Sales DESC
LIMIT 1;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[6]:

	Month	Total_Sales
0	2025-12	131135.53

Key Insight

December 2025 was the strongest month, generating $131,135.53 in total sales.

Promotions Analysis

This section summarizes promotion usage and discount amounts during the analyzed period.

Total Discount Amount

The following SQL query calculates the total discount amount generated by promotions.

In [7]:

```
# Execute the SQL query for total discount amount

query = """
SELECT
    ROUND(SUM("Discount Amount"), 2) AS Total_Discount_Amount
FROM promotions;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[7]:

	Total_Discount_Amount
0	90189.53

Key Insight

Promotions generated a total discount amount of $90,189.53 during the analyzed period.

Most Used Promotions

The following SQL query identifies the ten promotions with the highest number of uses.

In [8]:
```
# Execute the SQL query for the most used promotions

query = """
SELECT
    "Promotion",
    SUM("Uses") AS Total_Uses
FROM promotions
GROUP BY "Promotion"
ORDER BY Total_Uses DESC
LIMIT 10;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[8]:

	Promotion	Total_Uses
0	3x2 SIEMPRE	3823
1	10% GRANDE TABLE	2438
2	100% PROPIETARIOS	541
3	CORTESIA AL PRODUCTO	351
4	15% DESC. EMPLEADOS	218
5	V&E&L FEBRERO 2026	159
6	50% MODO TASTY	141
7	15 % DESC. WHISKYS	138
8	10% DESCUENTO BARRA	121
9	DESC MODO TASTY 50%	88

Key Insight

`3x2 SIEMPRE` was the most frequently used promotion with 3,823 uses, followed by `10% GRANDE TABLE` with 2,438 uses.

Highest Discount Amount by Promotion

The following SQL query identifies the promotions that generated the highest total discount amounts.

In [9]:
```
# Execute the SQL query for promotions with the highest discount amounts

query = """
```

```
SELECT
    "Promotion",
    ROUND(SUM("Discount Amount"), 2) AS Total_Discount_Amount
FROM promotions
GROUP BY "Promotion"
ORDER BY Total_Discount_Amount DESC
LIMIT 10;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[9]:

	Promotion	Total_Discount_Amount
0	3x2 SIEMPRE	45045.70
1	10% GRANDE TABLE	21141.53
2	100% PROPIETARIOS	10052.87
3	50% MODO TASTY	4945.45
4	CORTESIA AL PRODUCTO	2072.96
5	15 % DESC. WHISKYS	1402.61
6	V&E&L FEBRERO 2026	1171.92
7	DESC MODO TASTY 50%	974.02
8	30% PROPIETARIOS	766.34
9	15% DESC. BOCATTO	619.61

Key Insight

`3x2 SIEMPRE` generated the highest total discount amount at $45,045.70, followedby`10$
21,141.53.

Payment Methods Analysis

This section summarizes sales and transaction activity across the different payment methods used during the analyzed period.

Sales by Payment Method

The following SQL query calculates total sales generated by each payment method.

In [11]:
```
# Execute the SQL query for sales by payment method

query = """
```

```
SELECT
    "Payment Method",
    ROUND(SUM("Sales"), 2) AS Total_Sales
FROM payment_methods
GROUP BY "Payment Method"
ORDER BY Total_Sales DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[11]:

	Payment Method	Total_Sales
0	TC. DINERS	395902.18
1	TC. VISA	353691.02
2	TD. VISA	274220.15
3	TC. MASTERCARD	257354.16
4	EFECTIVO	92799.98
5	TC.AMEX	44562.36
6	UBEREATS	39221.83
7	TC. DISCOVER	21414.99
8	RAPPI	19264.04
9	CUENTA X COBRAR	18002.04
10	PEDIDOS YA	9558.86
11	TRANSFERENCIA	8378.23
12	MERMAS	6224.97
13	CONSUMO INTERNO	5251.59
14	ORDEN DE CONSUMO	3489.39
15	FACT REPROCESADA	1439.50
16	CANJE SCALA	941.62
17	CANJE	240.39
18	PAGO CON GIFT CARD	124.45
19	CHEQUE	92.64
20	MANUAL	11.16
21	BENDO	0.00

Key Insight

`TC. DINERS` generated the highest total sales at $395, 902.18, followedby`TC. VISA`with$353,691.02 and `TD. VISA` with $274,220.15.

Transactions by Payment Method

The following SQL query calculates the total number of transactions for each payment method.

In [12]:
```
# Execute the SQL query for transactions by payment method

query = """
SELECT
    "Payment Method",
    SUM("Transactions") AS Total_Transactions
FROM payment_methods
GROUP BY "Payment Method"
ORDER BY Total_Transactions DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[12]:

	Payment Method	Total_Transactions
0	TC. DINERS	6386
1	TC. VISA	5501
2	TD. VISA	4848
3	TC. MASTERCARD	4037
4	EFECTIVO	2565
5	UBEREATS	1315
6	RAPPI	784
7	TC.AMEX	638
8	MERMAS	617
9	MANUAL	595
10	CONSUMO INTERNO	348
11	TC. DISCOVER	347
12	PEDIDOS YA	332
13	TRANSFERENCIA	113
14	CUENTA X COBRAR	64
15	ORDEN DE CONSUMO	37
16	FACT REPROCESADA	13
17	PAGO CON GIFT CARD	4
18	BENDO	4
19	CANJE	2
20	CHEQUE	1
21	CANJE SCALA	1

Key Insight

`TC. DINERS` recorded the highest number of transactions with 6,386, followed by `TC. VISA` with 5,501 and `TD. VISA` with 4,848 transactions.

Monthly Sales by Payment Method

The following SQL query summarizes total sales by payment method for each month.

In [13]:
```
# Execute the SQL query for monthly sales by payment method
```

```python
query = """
SELECT
    strftime('%Y-%m', "Date") AS Month,
    "Payment Method",
    ROUND(SUM("Sales"), 2) AS Total_Sales
FROM payment_methods
GROUP BY Month, "Payment Method"
ORDER BY Month, Total_Sales DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[13]:

	Month	Payment Method	Total_Sales
0	2025-03	TC. DINERS	34480.09
1	2025-03	TC. VISA	24549.31
2	2025-03	TD. VISA	20143.59
3	2025-03	TC. MASTERCARD	19137.92
4	2025-03	EFECTIVO	7957.73
...	...	...	...
222	2026-03	CONSUMO INTERNO	288.33
223	2026-03	FACT REPROCESADA	189.99
224	2026-03	ORDEN DE CONSUMO	160.19
225	2026-03	TRANSFERENCIA	146.79
226	2026-03	MANUAL	0.00

227 rows × 3 columns

Key Insight

Payment method performance varied across the analyzed months, with card payments consistently representing the strongest sales contribution.

Employees Analysis

This section summarizes employee performance based on total sales and quantity sold during the analyzed period.

Employee Sales Ranking

The following SQL query ranks employees by total sales.

In [14]:

```python
# Execute the SQL query for employee sales ranking

query = """
SELECT
    "Employee",
    ROUND("Sales", 2) AS Total_Sales
FROM employees
ORDER BY Total_Sales DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[14]:

	Employee	Total_Sales
0	Employee 1	61641.63
1	Employee 2	54125.31
2	Employee 3	45762.89
3	Employee 4	42810.66
4	Employee 5	40654.99
5	Employee 6	24884.01
6	Employee 7	20890.97
7	Employee 8	8458.35
8	Employee 9	5343.32
9	Employee 10	4883.78
10	Employee 11	2091.95
11	Employee 12	1842.48
12	Employee 13	1282.71
13	Employee 14	576.60
14	Employee 15	453.18
15	Employee 16	397.66
16	Employee 17	73.21

Key Insight

`Employee 1` generated the highest total sales at
$61,641.63, followedby`Employee2`with$54,125.31 and `Employee 3` with $45,762.89.

Employee Quantity Ranking

The following SQL query ranks employees by total quantity sold.

In [15]:
```
# Execute the SQL query for employee quantity ranking

query = """
SELECT
    "Employee",
    ROUND("Quantity", 2) AS Total_Quantity
FROM employees
ORDER BY Total_Quantity DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[15]:

	Employee	Total_Quantity
0	Employee 1	15786.20
1	Employee 2	14256.90
2	Employee 3	12638.87
3	Employee 4	11382.50
4	Employee 5	10697.50
5	Employee 7	5788.33
6	Employee 6	3617.62
7	Employee 8	2264.91
8	Employee 9	1394.33
9	Employee 10	1305.00
10	Employee 11	469.00
11	Employee 12	385.00
12	Employee 13	331.00
13	Employee 14	162.60
14	Employee 15	116.40
15	Employee 16	107.00
16	Employee 17	23.00

Key Insight

`Employee 1` recorded the highest total quantity sold with 15,786.20 units, followed by `Employee 2` with 14,256.90 and `Employee 3` with 12,638.87 units.

Products Analysis

This section summarizes product performance based on sales, quantity sold, category contribution, and average sales value per unit.

Top Products by Sales

The following SQL query identifies the fifteen products with the highest total sales.

In [16]:
```
# Execute the SQL query for top products by sales

query = """
SELECT
    "Product",
    ROUND("Sales", 2) AS Total_Sales
FROM products
ORDER BY Total_Sales DESC
LIMIT 15;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[16]:

	Product	Total_Sales
0	Lomo Fino A La Parrilla	9538.15
1	Pasta Favorita Del Chef	7966.39
2	Salmone	7405.87
3	Chicken Parmegiana	6467.60
4	Lasagna	6465.16
5	Carbonara	6334.68
6	Aperol Spritz	6223.70
7	Risotto Al Lomo Saltado	6093.58
8	Picana Angus Importada	6076.00
9	Pulpo a la Parrilla	6073.65
10	Pollo A La Parrilla	6054.75
11	New Italian Lunch	5263.20
12	Ravioli Di Ricotta/Esp	5096.16
13	Grilled Steak	4819.62
14	Limonada Frutos Rojos	4665.96

Key Insight

`Lomo Fino A La Parrilla` was the highest-selling product, generating $9,538.15 in total sales, followed by `Pasta Favorita Del Chef` with $7,966.39 and `Salmone` with $7,405.87.

Top Products by Quantity Sold

The following SQL query identifies the fifteen products with the highest quantity sold.

In [17]:

```
# Execute the SQL query for top products by quantity sold

query = """
SELECT
    "Product",
    ROUND("Quantity", 2) AS Total_Quantity
FROM products
ORDER BY Total_Quantity DESC
LIMIT 15;
"""

result = pd.read_sql_query(query, conn)
```

```
result
```

Out[17]:

	Product	Total_Quantity
0	3 X 2 Siempre	1930.0
1	Agua Sin Gas San Felipe	1734.0
2	Limonada Frutos Rojos	1464.0
3	Guitig Clasica	1244.0
4	Guitig	902.0
5	Limonada De Mandarina	811.0
6	Empaque Delivery	811.0
7	Coca Zero 300CC	809.0
8	Limonada Con H.B	793.0
9	Limonada Imperial	775.0
10	Coca Cola 300CC	765.0
11	Limonada	710.0
12	Guitig (Desec)	623.0
13	Aperol Spritz	524.0
14	Americano	503.0

Key Insight

`3 X 2 Siempre` recorded the highest quantity with 1,930 units, followed by `Agua Sin Gas San Felipe` with 1,734 units and `Limonada Frutos Rojos` with 1,464 units.

Several of the highest-volume products were beverages, promotional items, or operational items, showing that high quantity does not necessarily correspond to high revenue.

Sales by Product Category

The following SQL query calculates total sales for each product category.

In [18]:

```
# Execute the SQL query for sales by product category

query = """
SELECT
    "Category",
    ROUND(SUM("Sales"), 2) AS Total_Sales
FROM products
GROUP BY "Category"
```

```
ORDER BY Total_Sales DESC;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[18]:

	Category	Total_Sales
0	Pasta	51609.07
1	Meat, chicken & seafood	45586.08
2	3x2 Promotion	37675.42
3	Entrees	25276.99
4	Specials	23341.59
5	Juice	19103.62
6	Pizza & Calzone	18354.51
7	Risotto	13655.13
8	Salads	10107.59
9	Water	7667.81
10	Extras	6926.45
11	Soups	6882.70
12	Italian Lunch	6591.90
13	Dessert	6406.27
14	Wine	6005.48
15	Pop Beverages	4767.40
16	Coffee & Tea	4665.11
17	Events Menus	4596.90
18	Children	4056.28
19	Liquor	2047.97
20	Beer	2042.80

Key Insight

`Pasta` generated the highest total sales at 51,609.07, followed by `Meat, chicken & seafood` with 45,586.08 and `3x2 Promotion` with $37,675.42.

Average Sales Value per Unit

The following SQL query identifies products with the highest average sales value per unit among products with at least 20 units sold.

In [19]:
```
# Execute the SQL query for average sales value per unit

query = """
SELECT
    "Product",
    ROUND("Sales" / "Quantity", 2) AS Average_Sales_Per_Unit
FROM products
WHERE "Quantity" >= 20
ORDER BY Average_Sales_Per_Unit DESC
LIMIT 15;
"""

result = pd.read_sql_query(query, conn)

result
```

Out[19]:

	Product	Average_Sales_Per_Unit
0	Rack De Cordero Importado	36.68
1	Bife De Chorizo	34.92
2	Picana Angus Importada	27.74
3	Risotto Frutti Di Mare	25.50
4	Asap	24.90
5	Pasta Frutti Di Mare	24.05
6	Pesca A La Parrilla	23.72
7	Risotto Al Lomo Saltado	23.26
8	Lomo Saltado Huancaina	23.20
9	Salmone	23.14
10	Salmone Alla Parmigiana	23.00
11	Crispy Salmone	22.98
12	Pesca Torino	21.60
13	Ossobuco De La Nonna	21.48
14	Lomo Blanco Y Negro	21.33

Key Insight

Among products with at least 20 units sold, `Rack De Cordero Importado` recorded the highest average sales value per unit at $36.68, followedby‘BifeDeChorizo‘at$34.92 and `Picana Angus Importada` at $27.74.

Average sales value per unit should not be interpreted as profit margin, since product costs are not included in the dataset.

Final Conclusions

The SQL analysis confirmed the main business patterns identified across the restaurant datasets.

Sales were strongly concentrated in the Food category, while December 2025 recorded the highest monthly sales during the analyzed period.

Promotional activity was dominated by `3x2 SIEMPRE`, both in usage and total discount amount.

Card payment methods generated the highest sales and transaction volumes, with `TC. DINERS` leading both metrics.

Employee performance showed a clear concentration among the top performers, with `Employee 1` ranking first in both sales and quantity sold.

Product analysis showed that the highest-selling products were not always the highest-volume products. Premium dishes generated higher sales value per unit, while several beverages and promotional items recorded high quantities with comparatively lower revenue.

Overall, SQL provided a structured and reproducible way to summarize, rank, filter, and compare the main business metrics across the restaurant datasets.

Author

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